

IDC INDIAN
DATA
CLUB

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21 DAYS SQL CHALLENGE

CHALLENGE STARTS FROM

3RD NOVEMBER 2025

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#SQLWithIDC

Day 18 (24/11): UNION and UNION ALL

🎯 Objective

To combine data from multiple tables or queries into a single result set and understand the difference between UNION and UNION ALL.

🔍 Topics Covered

- Using **UNION** to combine results without duplicates
- Using **UNION ALL** to keep all rows
- Aligning columns across multiple **SELECT** queries
- Ordering and filtering combined results

UNION

- Combines results from two queries
- Removes duplicate rows

```
SELECT column_list FROM table1  
UNION  
SELECT column_list FROM table2;
```

UNION ALL

- Combines results from two queries
- Keeps duplicates
- Faster than UNION

```
SELECT column_list FROM table1  
UNION ALL  
SELECT column_list FROM table2;
```

Practice Questions:

```
-- Combine patient names and staff names into a single list.  
SELECT name AS combined_name  
FROM patients  
UNION ALL  
SELECT staff_name  
FROM staff;
```

combined_name
Larry Dixon
Victor Taylor
Amber Obrien
Victoria Larson
Corey Whitaker
Thomas Atkins
Steven Howard
Chris Velazquez
Angela Lin
Melissa Mcfarland
Jordan Williams
Michelle Davis DVM
Dean Donovan
George Harper
Michael Wang
Scott Alexander
Jake Shaw
Bradley Johnson DDS
Alexander Gomez
Donald Medina
Michael Morrison

Practice Questions:

```
-- Create a union of
-- high satisfaction patients (>90) and low satisfaction patients (<50).
```

```
SELECT
```

```
    patient_id,
    name,
    age,
    arrival_date,
    departure_date,
    service,
    satisfaction,
    'High Satisfaction' AS satisfaction_label
```

```
FROM patients
```

```
WHERE satisfaction > 90
```

```
UNION
```

```
SELECT
```

```
    patient_id,
    name,
    age,
    arrival_date,
    departure_date,
    service,
    satisfaction,
    'Low Satisfaction' AS satisfaction_label
```

```
FROM patients
```

```
WHERE satisfaction < 50;
```

patient_id	name	age	arrival_date	departure_date	service	satisfaction	satisfaction_label
PAT-029113eb	Victoria Larson	84	2025-12-22	2025-12-31	surgery	97	High Satisfaction
PAT-03ae75aa	Jordan Williams	61	2025-08-27	2025-08-30	surgery	97	High Satisfaction
PAT-04be60be	Bradley Johnson DDS	83	2025-11-18	2025-12-02	general_medicine	93	High Satisfaction
PAT-07a9ee88	Michael Morrison	12	2025-08-17	2025-08-30	surgery	95	High Satisfaction
PAT-0b1acdd9	Aaron Schroeder	1	2025-02-27	2025-03-10	emergency	97	High Satisfaction
PAT-0c687528	Monica Ellis	39	2025-04-24	2025-05-06	ICU	96	High Satisfaction
PAT-0f24de9a	Tammy Mcdowell	50	2025-02-11	2025-02-13	general_medicine	91	High Satisfaction
PAT-1056838c	Sandra Drake	49	2025-04-26	2025-05-10	general_medicine	94	High Satisfaction
PAT-1112a754	Ashley Smith	35	2025-05-27	2025-06-08	surgery	99	High Satisfaction
PAT-1321d568	Mr. Lawrence Edwards	39	2025-10-13	2025-10-22	emergency	92	High Satisfaction
PAT-14171b1d	Gary Wells	55	2025-05-21	2025-06-03	emergency	97	High Satisfaction
PAT-15367ce5	Monica Miller	59	2025-05-18	2025-05-21	general_medicine	91	High Satisfaction
PAT-15b8abcc	Travis Tucker	29	2025-04-18	2025-04-27	general_medicine	96	High Satisfaction
PAT-15c81fbe	Holly Farmer	11	2025-02-18	2025-02-28	general_medicine	97	High Satisfaction
PAT-17ceaafc	Mary Miller	54	2025-04-01	2025-04-07	ICU	96	High Satisfaction
PAT-1aacdb2d	James Gilbert	61	2025-06-07	2025-06-19	emergency	94	High Satisfaction
PAT-1b2fe77a	Janet Kane DVM	89	2025-07-03	2025-07-15	emergency	93	High Satisfaction
PAT-1bc22a80	David Alvarez	8	2025-02-22	2025-03-06	surgery	94	High Satisfaction
PAT-1c834d8e	Victor Brown	35	2025-03-14	2025-03-22	general_medicine	92	High Satisfaction
PAT-1f02565c	Tina Hall	1	2025-01-22	2025-01-25	ICU	96	High Satisfaction
PAT-1fd2e6ec	Joseph Cooper	78	2025-01-22	2025-01-28	emergency	94	High Satisfaction

Practice Questions:

```
-- List all unique names from both patients and staff tables.  
SELECT name AS combined_name  
FROM patients  
UNION  
SELECT staff_name  
FROM staff;
```

combined_name
Larry Dixon
Victor Taylor
Amber Obrien
Victoria Larson
Corey Whitaker
Thomas Atkins
Steven Howard
Chris Velazquez
Angela Lin
Melissa Mcfarland
Jordan Williams
Michelle Davis DVM
Dean Donovan
George Harper
Michael Wang
Scott Alexander
Jake Shaw
Bradley Johnson DDS
Alexander Gomez
Donald Medina
Michael Morrison

Daily Challenge:

```
-- Create a comprehensive personnel and patient list showing:
-- identifier (patient_id or staff_id), full name, type ('Patient' or 'Staff'),
-- and associated service. Include only those in 'surgery' or 'emergency' services.
-- Order by type, then service, then name.
```

```
SELECT
    patient_id AS identifier,
    name AS full_name,
    'Patient' AS type,
    service
FROM patients
WHERE service IN ('surgery', 'emergency')
UNION ALL
SELECT
    staff_id AS identifier,
    staff_name AS full_name,
    'Staff' AS type,
    service
FROM staff
WHERE service IN ('surgery', 'emergency')
ORDER BY type, service, full_name;
```

identifier	full_name	type	service
PAT-0f8876bf	Aaron Parker	Patient	emergency
PAT-0b1acdd9	Aaron Schroeder	Patient	emergency
PAT-09edc78a	Adam Jackson	Patient	emergency
PAT-0c7c720b	Aimee Turner	Patient	emergency
PAT-8b68dd5f	Alex Hernandez	Patient	emergency
PAT-d8b5f75c	Alexander Brown	Patient	emergency
PAT-06a8ea25	Alexander Gomez	Patient	emergency
PAT-69b43c9f	Allen Rosales	Patient	emergency
PAT-de5fbcb3	Allison Smith	Patient	emergency
PAT-73381b12	Alyssa Long	Patient	emergency
PAT-8bfedd28	Amanda Harvey	Patient	emergency
PAT-a8b5abcb	Amanda Hill	Patient	emergency
PAT-3ca8e0e7	Amanda Terrell	Patient	emergency
PAT-a01bf534	Amber Vang	Patient	emergency
PAT-85f8f76d	Amy Garcia	Patient	emergency
PAT-686bfae7	Amy Martinez	Patient	emergency
PAT-532a2fa4	Andrew Avila	Patient	emergency
PAT-7472f19a	Andrew Harper	Patient	emergency
PAT-6813152c	Angel May	Patient	emergency
PAT-9da5da0c	Angela Vaughn	Patient	emergency
PAT-6527c233	Angelica Keith	Patient	emergency